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Summary

Caltech Economics PhD student working across machine learning and economic theory. Build end-to-end pipelines using large-scale data. Job market paper introduces a novel NLP architecture for patent-litigation prediction.

Education

2021–2027 (exp.) **Ph.D. in Quantitative Economics and Econometrics**, *California Institute of Technology (Caltech)*, Pasadena, CA, USA
Selected coursework: Machine Learning, Deep Learning, Advanced Econometrics

2017–2020 **M.Sc. in Economics and Social Sciences**, *Bocconi University*, Milan, Italy
Bocconi Graduate Merit Award

2012–2016 **B.A. in Business and Management**, *National Economics University of Vietnam*
Merit awards for performance in the top 5% of students (2013, 2014, 2015)

Selected Research

Multi-Text Representation Learning for Patent Validity Prediction
Job Market Paper

Abstract: Predicting patent validity litigation outcomes requires combining information from documents that serve distinct roles and may depend on one another for interpretation. Using a newly constructed dataset of UK patent litigation, I investigate two main questions: how to represent texts whose meaning depends on other texts, and how to aggregate information across multiple challenges into an overall validity prediction. To characterize textual dependencies, I define maximum dependency depth as the length of the longest chain of dependencies. I then construct expanded text representations that explicitly incorporate relevant content from the texts on which they depend. To aggregate information across challenges, I develop a multiple-instance learning architecture tailored to patent validity litigation. The architecture uses cross-attention, constructing queries from representations of the challenged patent claim and keys and values from representations of the texts cited against it. Empirical results show that the proposed architecture performs better than the architectures used in the literature as well as ChatGPT.

Optimal Stopping Problems with Multiple Agents
Single-authored; accepted for poster presentation at ACM EC.

Abstract: We study an optimal stopping problem between a principal and a group of experts in an environment without transfers. The principal must decide when to take an action without knowing its payoff but can consult a group of experts. Each expert derives a constant utility from the action and privately observes information about its payoff to the principal. A conflict of interest arises: the principal wishes to act only when it is worthwhile, whereas experts prefer the action to be taken as early as possible. We characterize the optimal stopping rule based on the reports of the experts and show that the optimal mechanism is monotone, consisting of a finite sequence of stages of finite length. In each stage, the principal stops if and only if at least a threshold number of experts recommend stopping, with the threshold decreasing across stages and becoming zero in the final stage. Finally, we derive the principal's asymptotic payoff as the number of experts grows large.

Personal Projects

Do Reviews Affect Stock Performance? Evidence from the Entertainment Industry

- Studied whether critic and user reviews predict stock returns for Japanese and U.S. video game companies.
- Built Scrapy pipelines to collect critic and user reviews from Metacritic and used Python, pandas, and SQL to

construct daily review-based signals aligned with financial-market data.
- An event-driven portfolio strategy generated about 11.29% annualized Fama–French five-factor alpha in the study sample, with a t-statistic of 3.91, a Sharpe ratio of 1.1, and a maximum drawdown of around 12.52%.
- The strategy remained robust across alternative holding horizons, rebalancing frequencies, and signal thresholds.

Research Experience

Research Assistant — Bocconi University
Market Microstructure (Dark Pools)

- Built and analyzed a game-theoretic model in Mathematica to study the introduction of dark pools, showing that their addition increases market depth and reduces spreads.
- Specified model primitives and the equilibrium concept, implemented equilibrium computation, and conducted comparative analyses across market structures.

Technical Skills

Programming	Python (PyTorch, TensorFlow, pandas, NumPy, scikit-learn, statsmodels, Scrapy, Playwright, Transformers), SQL, DuckDB, Mathematica
Machine learning	Supervised learning, neural networks, NLP/LLMs, pretrained embeddings, fine-tuning and LoRA, classical ML methods (logistic regression, SVMs, tree-based models), and model ensembling
Data engineering	Built large-scale web scraping and data-processing pipelines using Scrapy, Playwright, pandas, NumPy, SQL, and parallel processing; worked with both structured data and unstructured text
Empirical methods	Event studies, portfolio backtesting, econometrics, feature engineering
Theory & optimization	Dynamic programming, convex optimization, integer programming, game theory, mechanism design